

Perceptron Convergence Theorem: A Step-by-Step Proof

Setup. Let $\mathcal{D} = \{(x_p, t_p)\}_{p=1}^P$ with $x_p \in \mathbb{R}^d$ augmented by a 1 to absorb the bias, and labels $t_p \in \{+1, -1\}$. Define

$$\mathcal{F}_+ := \{x_p : t_p = +1\}, \quad \mathcal{F}_- := \{x_p : t_p = -1\},$$

and the transformed set $\mathcal{F} := \mathcal{F}_+ \cup \{-x : x \in \mathcal{F}_-\}$ whose elements all have target +1. Perceptron updates with learning rate $\eta = 1$ are

$$\text{if } \langle x, w \rangle \leq 0 \text{ then } w \leftarrow w + x, \quad \text{else no change.}$$

Theorem 1 (Perceptron Convergence). *If there exists $w^* \in \mathbb{R}^{d+1}$ such that $\langle x, w^* \rangle > 0$ for all $x \in \mathcal{F}$, then the perceptron makes a finite number of mistakes and halts. Moreover, if we scale w^* so that $\|w^*\| = 1$, define*

$$m := \min_{x \in \mathcal{F}} \langle x, w^* \rangle \quad (> 0), \quad M := \max_{x \in \mathcal{F}} \|x\|^2,$$

then the total number of updates K satisfies $K \leq M/m^2$.

Proof. We prove two inequalities: a lower bound on the projection $\langle w_k, w^* \rangle$ that grows linearly in k , and an upper bound on $\|w_k\|^2$ that grows at most linearly in k . Combining them yields a finite mistake bound.

Step 1: Notation for mistake sequence. Index the updates by $k = 1, 2, \dots$ in the order they occur. Let $x_{k-1} \in \mathcal{F}$ be the example that caused the $(k-1)$ -st mistake, and let w_k be the weight vector after applying that update. With $w_0 = \mathbf{0}$ (WLOG), the update rule is

$$w_k = w_{k-1} + x_{k-1}, \quad \text{with } \langle x_{k-1}, w_{k-1} \rangle \leq 0.$$

By unrolling,

$$w_k = \sum_{i=0}^{k-1} x_i.$$

Step 2: Linear growth of the projection on w^* . For any mistake x_{k-1} we have $\langle x_{k-1}, w^* \rangle \geq m$ by definition of m . Hence

$$\langle w_k, w^* \rangle = \langle w_{k-1} + x_{k-1}, w^* \rangle = \langle w_{k-1}, w^* \rangle + \langle x_{k-1}, w^* \rangle \geq \langle w_{k-1}, w^* \rangle + m.$$

By induction,

$$\langle w_k, w^* \rangle \geq k m. \tag{1}$$

Step 3: Quadratic upper bound via norm growth. Compute the squared norm after a mistake:

$$\|w_k\|^2 = \|w_{k-1} + x_{k-1}\|^2 = \|w_{k-1}\|^2 + 2 \langle x_{k-1}, w_{k-1} \rangle + \|x_{k-1}\|^2 \leq \|w_{k-1}\|^2 + \|x_{k-1}\|^2,$$

since $\langle x_{k-1}, w_{k-1} \rangle \leq 0$ at a mistake. Iterating and using $\|x_i\|^2 \leq M$ for all i gives

$$\|w_k\|^2 \leq \sum_{i=0}^{k-1} \|x_i\|^2 \leq k M. \quad (2)$$

Step 4: Combine (1) and Cauchy–Schwarz with (2). By Cauchy–Schwarz,

$$\langle w_k, w^* \rangle \leq \|w_k\| \|w^*\| = \|w_k\| \quad \text{since } \|w^*\| = 1.$$

Together with (1) and (2),

$$k m \leq \langle w_k, w^* \rangle \leq \|w_k\| \leq \sqrt{k M}.$$

Therefore $km \leq \sqrt{kM}$, i.e. $k \leq M/m^2$. Hence only finitely many mistakes occur, so the algorithm halts. $\square$

Remarks. (i) If one instead defines a margin $\gamma := \min_p t_p \langle x_p, \hat{w}^* \rangle$ for a unit vector $\hat{w}^*$, and $R := \max_p \|x_p\|$, then $K \leq (R/\gamma)^2$. (ii) Scaling w^* to unit norm converts (M, m) to (R^2, γ) .

1 Concrete matrices, vectors, and an update trace

1.1 Dataset as matrices (augmented bias)

Let $d = 2$ features and bias absorbed by augmentation. Take four separable points:

$$\tilde{x}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, \quad \tilde{x}_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, \quad \tilde{x}_3 = \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}, \quad \tilde{x}_4 = \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \quad y = \begin{bmatrix} +1 \\ +1 \\ -1 \\ -1 \end{bmatrix}.$$

Row-stacked design matrix:

$$X_{\text{rows}} = \begin{bmatrix} 2 & 2 & 1 \\ 3 & 1 & 1 \\ -1 & -2 & 1 \\ -2 & -1 & 1 \end{bmatrix} \in \mathbb{R}^{4 \times 3}, \quad \text{where row } p \text{ equals } \tilde{x}_p^\top.$$

Column-stacked view (useful algebraically):

$$X_{\text{cols}} = [\tilde{x}_1 \quad \tilde{x}_2 \quad \tilde{x}_3 \quad \tilde{x}_4] \in \mathbb{R}^{3 \times 4}.$$

Standard perceptron update (labeled form). With labels $y_p \in \{\pm 1\}$ and $w \in \mathbb{R}^3$,

$$\text{if } y_p \langle w, \tilde{x}_p \rangle \leq 0 \implies w \leftarrow w + y_p \tilde{x}_p.$$

Equivalently, over a sequence of mistakes at indices $p_0, \dots, p_{k-1}$,

$$w_k = \sum_{i=0}^{k-1} y_{p_i} \tilde{x}_{p_i} \quad (\text{so } w_k = X_{\text{cols}} \alpha_k \text{ for some } \alpha_k \in \mathbb{N}_0^4 \text{ counting mistakes per example}).$$

1.2 Transformed set $\mathcal{F}$ used in the proof

Flip the negatives so every element has target +1:

$$\mathcal{F} = \left\{ \tilde{x}_1, \tilde{x}_2, -\tilde{x}_3 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}, -\tilde{x}_4 = \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix} \right\}.$$

In this view the update is: if $\langle w, x \rangle \leq 0$ for some $x \in \mathcal{F}$ then $w \leftarrow w + x$.

1.3 A full pass: explicit mistake trace

Initialize $w_0 = \mathbf{0}$. Present the four examples in order $(1, 2, 3, 4)$.

Step 1 ($p = 1$): $y_1 \langle w_0, \tilde{x}_1 \rangle = 0 \leq 0 \Rightarrow w_1 = w_0 + y_1 \tilde{x}_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}.$

Step 2 ($p = 2$): $y_2 \langle w_1, \tilde{x}_2 \rangle = (+1)(2 \cdot 3 + 2 \cdot 1 + 1 \cdot 1) = 9 > 0 \Rightarrow w_2 = w_1.$

Step 3 ($p = 3$): $y_3 \langle w_2, \tilde{x}_3 \rangle = (-1)(2 \cdot (-1) + 2 \cdot (-2) + 1 \cdot 1) = (-1)(-5) = 5 > 0 \Rightarrow w_3 = w_2.$

Step 4 ($p = 4$): $y_4 \langle w_3, \tilde{x}_4 \rangle = (-1)(2 \cdot (-2) + 2 \cdot (-1) + 1 \cdot 1) = (-1)(-5) = 5 > 0 \Rightarrow w_4 = w_3.$

All points are correctly classified after a single update: $K = 1$.

1.4 A separating hyperplane and the numeric bound

A simple separator is $w^* = (1, 1, -1)$. For all p ,

$$y_p \langle w^*, \tilde{x}_p \rangle \in \{3, 4\} > 0 \quad \Rightarrow \quad \text{separable.}$$

Margin form: set $\hat{w}^* = w^* / \|w^*\|$ with $\|w^*\| = \sqrt{3}$ and

$$\gamma := \min_p y_p \langle \hat{w}^*, \tilde{x}_p \rangle = \frac{3}{\sqrt{3}} = \sqrt{3}.$$

Radius $R := \max_p \|\tilde{x}_p\| = \sqrt{11}$. Hence the classic bound

$$K \leq \left(\frac{R}{\gamma} \right)^2 = \frac{11}{3} < 4,$$

consistent with the observed $K = 1$.

1.5 Same example without augmentation (separate bias)

Let $x_p \in \mathbb{R}^2$, $w \in \mathbb{R}^2$, $b \in \mathbb{R}$. Update

$$\text{if } y_p (w^\top x_p + b) \leq 0 \quad \Rightarrow \quad \begin{cases} w \leftarrow w + y_p x_p, \\ b \leftarrow b + y_p. \end{cases}$$

This is algebraically identical to the augmented rule via $\tilde{x}_p = (x_p, 1)$ and $\tilde{w} = (w, b)$.

Geometric intuition. Each mistake adds a vector pointing “toward correctness”: $w_{k+1} = w_k + y_p \tilde{x}_p$ increases the projection $\langle w_{k+1}, \hat{w}^* \rangle$ by at least γ , while $\|w_{k+1}\|$ grows at most by $\|\tilde{x}_p\| \leq R$. Linear growth below and sublinear growth above force a finite K .

2 Kessler's construction for multiclass perceptron

2.1 Setup and Kessler lift

Let $K \geq 2$ classes, augmented inputs $\tilde{x} \in \mathbb{R}^{d+1}$, labels $y \in \{1, \dots, K\}$. Collect class weights as a matrix $W \in \mathbb{R}^{K \times (d+1)}$ with rows $w_c^\top$. Prediction is

$$\hat{y}(\tilde{x}) = \arg \max_{c \in [K]} \langle w_c, \tilde{x} \rangle.$$

Multiclass separability with margin. There exists W^* and $\gamma > 0$ such that for all $(\tilde{x}, y)$ and all $j \neq y$,

$$\langle w_y^* - w_j^*, \tilde{x} \rangle \geq \gamma, \quad R := \max_{(\tilde{x}, y)} \|\tilde{x}\| < \infty. \quad (\diamond)$$

Kessler lift. Define the block embedding $\Phi : \mathbb{R}^{d+1} \times [K] \rightarrow \mathbb{R}^{K(d+1)}$ by

$$\Phi(\tilde{x}, y) = \underbrace{(0, \dots, 0)}_{(y-1) \text{ blocks}}, \tilde{x}, \underbrace{(0, \dots, 0)}_{(K-y) \text{ blocks}},$$

and the contrast vector

$$\Delta\Phi(\tilde{x}, y, j) := \Phi(\tilde{x}, y) - \Phi(\tilde{x}, j), \quad j \neq y.$$

Vectorize W as $v := \text{vec}(W) \in \mathbb{R}^{K(d+1)}$ so that $\langle v, \Phi(\tilde{x}, c) \rangle = \langle w_c, \tilde{x} \rangle$. Then the margin constraints $(\diamond)$ are equivalent to

$$\langle v^*, \Delta\Phi(\tilde{x}, y, j) \rangle \geq \gamma \quad \forall j \neq y.$$

Note $\|\Delta\Phi(\tilde{x}, y, j)\|^2 = \|\tilde{x}\|^2 + \|\tilde{x}\|^2 = 2\|\tilde{x}\|^2 \leq 2R^2$.

2.2 Multiclass perceptron as binary perceptron in the lifted space

Given an example $(\tilde{x}, y)$, let $j = \arg \max_{c \neq y} \langle w_c, \tilde{x} \rangle$ be the most offending rival. On mistake $(\langle w_y, \tilde{x} \rangle \leq \langle w_j, \tilde{x} \rangle)$ perform

$$W_y \leftarrow W_y + \tilde{x}, \quad W_j \leftarrow W_j - \tilde{x} \iff v \leftarrow v + \Delta\Phi(\tilde{x}, y, j).$$

Thus the multiclass update equals a standard perceptron update on the lifted, always-positive examples $\Delta\Phi(\tilde{x}, y, j)$.

2.3 Convergence and mistake bound

Under $(\diamond)$, the binary perceptron proof applies verbatim in the lifted space. Let $v_0 = 0$ and index mistakes by k . Define

$$m := \min_{(\tilde{x}, y), j \neq y} \langle v^*, \Delta\Phi(\tilde{x}, y, j) \rangle = \gamma, \quad M := \max_{(\tilde{x}, y), j \neq y} \|\Delta\Phi(\tilde{x}, y, j)\|^2 \leq 2R^2.$$

Then after K mistakes,

$$K \leq \frac{M}{m^2} \leq \frac{2R^2}{\gamma^2}.$$

Equivalently, with Frobenius norm, $\|W^*\|_F = 1$ without loss of generality, the standard inequalities yield

$$\langle \text{vec}(W_K), \text{vec}(W^*) \rangle \geq K\gamma, \quad \|W_K\|_F^2 \leq K \cdot 2R^2,$$

and CauchySchwarz gives the same bound.

2.4 Concrete block form and a tiny trace

For $K = 3$ and $d = 2$, write $\tilde{x} = (x_1, x_2, 1)^\top$ and

$$W = \begin{bmatrix} w_{11} & w_{12} & b_1 \\ w_{21} & w_{22} & b_2 \\ w_{31} & w_{32} & b_3 \end{bmatrix}, \quad v = \text{vec}(W) = \begin{bmatrix} w_{11} \\ w_{12} \\ b_1 \\ w_{21} \\ w_{22} \\ b_2 \\ w_{31} \\ w_{32} \\ b_3 \end{bmatrix} \in \mathbb{R}^9.$$

For $(\tilde{x}, y = 2)$,

$$\Phi(\tilde{x}, 2) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ x_1 \\ x_2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \Delta\Phi(\tilde{x}, 2, 3) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ x_1 \\ x_2 \\ 1 \\ -x_1 \\ -x_2 \\ -1 \end{bmatrix}.$$

Suppose one mistake occurs on $(\tilde{x}, y = 2)$ predicted as $j = 3$. Then

$$W_2 \leftarrow W_2 + \tilde{x}, \quad W_3 \leftarrow W_3 - \tilde{x}, \quad W_1 \text{ unchanged}.$$

Numerically, let $\tilde{x} = (2, 1, 1)^\top$. With $W = 0$,

$$\text{scores } (\langle w_c, \tilde{x} \rangle) = (0, 0, 0) \Rightarrow \text{tie} \rightarrow j = 3, \quad W_2 \leftarrow (2, 1, 1), \quad W_3 \leftarrow (-2, -1, -1).$$

On the next pass a correctly labeled $(\tilde{x}', y = 2)$ with $\tilde{x}' = (0, 2, 1)^\top$ produces scores

$$(\langle w_1, \tilde{x}' \rangle, \langle w_2, \tilde{x}' \rangle, \langle w_3, \tilde{x}' \rangle) = (0, 0 \cdot 2 + 1 \cdot 2 + 1 \cdot 1 = 3, 0 \cdot (-2) + 2 \cdot (-1) + 1 \cdot (-1) = -3),$$

hence no update.

2.5 Geometric and intuitive reading

Each mistake adds the *difference* between the correct and rival class blocks. In weight space this pushes w_y toward $\tilde{x}$ and pulls w_j away from $\tilde{x}$. The projection on the optimal separator W^* increases by at least γ per mistake, while the Frobenius norm grows at most by $\sqrt{2}\|\tilde{x}\|$. As in the binary case, linear growth below versus sublinear growth above yields a finite mistake count bounded by $2R^2/\gamma^2$.

Remarks. (i) The update $W_y \leftarrow W_y + \tilde{x}$, $W_j \leftarrow W_j - \tilde{x}$ is identical to the structured perceptron and to the Crammer–Singer multiclass hinge in the separable limit. (ii) A one-vs-rest perceptron can also be analyzed, but Kesslers lift gives a single binary reduction with a tight $\sqrt{2}$ radius factor.